LICHUAN CITY DOMESTIC WASTE LANDFILL SITE BIOGAS POLLUTION TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT

Report ID	*A+SH_SYST_VCS_VAL_6424*
Project title	*Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project*
Project ID	*5313*
Crediting period	*05/09/2024 to 04/09/2034 (both days included)*
Original date of issue	*05/01/2025*
Most recent date of issue	*25/11/2025*
Version	*01.2*
VCS Standard Version	*VCS Standard version 4.7*
Client	*Henan BCCY Environmental Energy Co., Ltd.*

Prepared by	*LGAI Technological Center, S.A. (Applus+ Certification)*
Approved by	*LGAI Technological Center S.A. (Applus+ Certification)* *VVB Technical Manager: Ms. Karen Elizabeth Vega*
Work carried out by	*Lead Auditor / Technical Expert: Doris Dai* *Technical Reviewer/ Technical Expert: Simon Shen*

Summary:

LGAI Technological Center, S.A. (hereafter referred to as "Applus+ Certification") has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of project activity "Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project" (hereafter referred to as "the project activity") and reported in the PD.

The project activity is a landfill gas recovery and power generation project located at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China, of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2 MW consisting of 2 sets of 1,000 kW generators. The project uses LFG from Lichuan landfill site for power generation. During the 10-year fixed crediting period, the power generated by the project is expected to be 75,670 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 456,696 tCO_2e during the 10-year fixed crediting period.

The project activity is available at https://registry.verra.org/app/projectDetail/VCS/5313.

In the course of Validation, 00 Corrective Action Requests (CARs), 01 Clarification Requests (CLs) and no Forward action request (FARs) were raised.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- *AMS-III.G: Landfill methane recovery, version 10.0 /4/*
- *AMS-I.D: Grid connected renewable electricity generation, version 18.0 /4/*
- *Emissions from solid waste disposal sites, version 08.1 /8/*
- *Tool to calculate the emission factor for an electricity system, version 07.0 /7/*
- *Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0 /10/*
- *Positive lists of technologies, version 04.0 /11/*

The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide, version 4.4 /12/ and VCS Standard version 4.7 /13/.

- *CDM Validation and Verification Standard for project activities, version 03.0 /5/*
- *CDM Project Standard for project activities, version 03.0 /6/*
- *VCS Standard, version 4.7*
- *VCS Program Guide, version 4.4*

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated Verified Carbon Units (VCUs).

A risk-based approach has been followed to perform this validation activity. VVB applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per VCS Standard version 4.7 clause 4.1.10 (4). The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfilment of the stated criteria of VCS.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center, S.A. (hereafter referred to as "Applus+ Certification") has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of project activity "Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project" (hereafter referred to as "the project activity") and reported in the PD.

LGAI Technological Center, S.A. as the validation and verification body of the project activity has been accredited as a DOE by UNFCCC and also meets the competence requirements as set out in ISO 14065: 2020.

The objective of this validation is to ensure that reported information in the PD of "Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project" is complete and accurate in accordance with applicable VCS standards and relevant UNFCCC requirements.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the project design (PD). The PD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board, including the approved baseline and monitoring methodology AMS-III.G version 10.0 and AMS-I.D version 18.0. The validation was based on the requirements in the CDM Validation and Verification standard for project activities version 03.0 and VCS program guide version 4.4 and standard version 4.7.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the project proponent. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The only purpose of the validation is its usage during the registration/issuance processes as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

Applus+ Certification takes into consideration that the risk of material misstatements as part of the performance of a VCS validation assessment varies with factor/elements such as (i) the complexity and subjectivity associated with the process; (ii) the availability and reliability of relevant data; (iii) quality and quantity of information and evidences; (iv) the number and significance of assumptions that are made and finally, (v) the degree of uncertainty associated with such factors/elements.

In the context of the performance of the VCS validation assessment vis-à-vis considered assumptions, under conformance with Applus+ Certification working procedures, the validation team appointed by Applus+ Certification is required to perform an evaluation/test of the overall reasonableness, completeness and accuracy of all relevant assumptions used for the description of the design of the proposed VCS project activity, the project start date, the delineation of the project boundary, the identification of the baseline scenario, assessment and demonstration of additionality, applied GHG calculation approaches and the design of the monitoring plan (based on ex-ante determined parameters and parameters to be monitored ex-post).

Such reasonability analysis is performed by the Applus+ Certification's validation team (that holds significant experience and expertise with project-based initiatives encompassing collection and destruction/utilization of LFG) by assuming related limitations and by inter-alia taking into consideration the following relevant aspects:

- *Regional, national and international practice related to GHG mitigation/abatement measure and technology applied by the proposed VCS project activity.*
- *Definitions, concepts and principles of the applied baseline and monitoring methodology and methodological tools (which are assumed as being designed by being rooted in sound and widely accepted scientific principles and knowledge).*
- *Overall credibility of quality and reliability of data, information, facts and evidences used inter-alia for the identification of baseline scenario and baseline emissions (with consideration of contrary or inconsistent data, information and evidence where applicable).*
- *Relevance and credibility of identified stakeholders for the proposed VCS project activity.*
- *Regulatory, legal and operational requirements related to the measure and technologies encompassed by the proposed VCS project activity.*

Also, part of its validation assessment, a risk assessment is therefore performed as part of the validation planning in order to identify (and mitigate, if applicable) potential factors and/or elements that may negatively influence the credibility and fairness of the outcomes of the assessment. In case significant risks are identified, further tests may have been considered as required in order to evaluate the reasonableness of assumptions (and/or alternative assumptions) or outcomes, if applicable.

The validation report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. The assessment team applied a materiality threshold of 5% (for projects with ER less than or equal to 300,000 tCO_2e per year) with respect to emission or misstatements concerning reported quantities as per para 3.10.1 and 4.1.10 (4) of VCS Standard version 4.7.

Finally, it is also crucial to note that, as part of the technical review of the assessment, the reasonableness of the identification and eventual mitigation of such risks (as well as other relevant judgements and decisions made by the assessment team) is to be confirmed by the appointed review team (which is independent to the validation team).

In conclusion, Applus+ Certification planned and performed the Validation by obtaining objective evidence that substantiate the reasonableness of PP's assumptions and determine whether the project and its GHG statement conforms with the regulatory documents, as well as evaluates the limitations and methods that support a claim about the outcome of future activities, in accordance with the VCS Standard 4.7.

1.4 Summary Description of the Project

Project title	*Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project*
VCS reference number	*5313*
Project Proponent	*Henan BCCY Environmental Energy Co., Ltd. (Project Proponent, host country, P. R. China)*
Location of the project	*Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China* *Geographic coordinates:* *Longitude: 108° 58′ 53.205″ E, Latitude: 30° 18′ 1.403″ N*
Project start date	*Construction date: 20/04/2024* *Operation start date: 05/09/2024*
Version of VCS PD	*Version 04 dated 24/11/2025*
Applied Methodology/Version	*AMS-III.G, version 10.0* *AMS-I.D, version 18.0*

Scope/Technical Area	*1/1.1* *13/13.1*

The project activity is a landfill gas recovery and power generation project located at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China, of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2 MW consisting of 2 sets of 1,000 kW generators. The project uses LFG from Lichuan landfill site for power generation. During the 10-year fixed crediting period, the power generated by the project is expected to be 75,670 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 456,696 tCO_2e during the 10-year fixed crediting period.

2 VALIDATION PROCESS

2.1 Method and Criteria

The validation process has been conducted following the regulatory documents and based on the ISO 14064-3: 2019 and ISO 14065: 2020 using standard auditing techniques as required by the VCS Standard version 4.7. The validation process has been conducted through an onsite inspection (see below), thus, no need for a risk analysis for remote inspections has been determined for this Validation (cf. VCS Standard version 4.7. Para 4.1.13).

No ICT means have been used for the conduction of this validation process other than those for the provision of documents and transferring information and undertake normal communications with the PP(s). Those ICT means supporting the process in this sense, have been used to an extent sufficient to cover the background necessities of the validation process and have demonstrated enough effectiveness to conduct it in accordance with the relevant applicable regulatory requirements. No virtual sites have been included in the validation process according to the definition in the IAF MD4: 2023 document. Instead, the documentation has been provided upon request from the VVB by other means (such as physical documents, mail, transfer applications, etc.).

Applus+ Certification completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable):

- Project Details;

- Application of Methodology;

- Estimated GHG Emission Reduction and Removals;

- Monitoring;

- Safeguards and SDG contributions etc.

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leakage if relevant;

- Protocols used to estimate or measure GHG emissions from these sources;

- Collection and handling of data;

- Controls on the collection and handling of data;

Based on the above, the VVB prepared a Validation Plan and applied an evidence-gathering plan designed to collect sufficient and appropriate evidence upon which to base the conclusion and to consider the inherent risk for the validation. The evidence-gathering plan is based on analytical procedures and tests and considers the thresholds for materiality as per the regulatory documents applied for this validation (5% as per the applied VCS Standard version 4.7).

The applied evidence-gathering plan is used by the VVB to determine the conformity of the *statements made by the client against the principles and requirements of the applied regulatory documents during this validation. The evidence-gathering plan designed by the VVB takes into consideration, as a minimum:*

- The selection and management of the data and information;

- The processes for collecting, processing, consolidating, aggregating and reporting the data and information;

- Systems and processes in place to ensure the validity and accuracy of the data and information;

- The design and maintenance of the system to control the data and information;

- Systems, processes and personnel that support system in place, including activities for ensuring data quality;

- The plans for instrument maintenance and calibration where appropriate;

Applus+ Certification validate that the reported information in the Project Description is complete and accurate in question. This involved an on-site audit and a desk review of the Project Design. This Validation Report describes the findings of this assessment. The process of validation involved the following relevant milestones:

- The signature of the contract for validation between the PP and the VVB;

- A kick-off meeting/communication to require the documentary material to perform an initial desk review;

- An initial desk review of the documentary evidences prepared by the Project Proponent;

- The issuance of findings coming from this initial desk review (if any);

- The conduction of an onsite audit (see Sections 2.3 and 2.4 below);

- The issuance of findings from the onsite audit and communication of deviations (if any) to the Project Proponent;

- The assessment of their resolution and closure, when appropriate;

- Once all the findings have been closed, the elaboration of this Validation Report, in which such findings (if any) are described;

- The conduction of an independent Technical Review;

- Upon conclusion of the Technical Review, the VVB may perform a final quality check before the issuance of the Final Validation Report (FVR) and signature of the corresponding Deed.

The process of this validation assessment has not implied any sampling plan.

The information of the assessment team is included in below:

Assessment team

According to the sectoral scopes / technical area and experiences in the sectoral or national *business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification). The composition of assessment team has to be approved by the LGAI Technological Center S.A. (Applus+ Certification) Central Site during its Contract Review stage, ensuring that the required skills are covered by the team. The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:*

- Leader Auditor (LA)

- Auditor (A)/ Auditor Trainee (AT)

- Technical Reviewer (TR)

- Technical Experts (TE)

- Other supporting roles such as Country Expert (CE) or Financial Expert (FE)

- Any of the above-mentioned roles in training (iT, e.g. AiT for auditor in training).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the assessment team as shown below:

Name	***Qualification***	***SS/TA Knowledge***	***Financial Expertise***	***Host country experience***	***Attendance to on-site visit***
Doris Dai	*LA/TE*	*Yes (1.1/13.1)*	*Yes*	*Yes*	*Yes*
Simon Shen	*TR/TE*	*Yes (1.1/13.1)*	*Yes*	*n/a*	*n/a*

***Doris Dai** (Master's Degree in Environmental Sciences, Bachelor's Degree in Environmental Technology) is an Auditor appointed by Applus+ Certification (LGAI Technological Center, S.A) for the GHG project assessment and auditing. She has more than 8 years of work experience in CDM/VCS project assessment and more than 3 years of working experience in GS4GG project assessment. Before she joined Applus+ LGAI, she has been working for CTI Certification as senior GHG Auditor for 3.5 years.*

***Simon Shen** (Master's Degree in Thermal Energy Engineering, Bachelor's Degree in Environmental Engineering) is an Auditor appointed by Applus+ LGAI for the GHG project assessment, auditing and technical review. He has more than 8 years of work experience in CDM/GS4GG/VCS project assessment and review with Applus+, apart from the years of experience working as GHG Auditor and ISO 9001/14001 in TUV SUD for 3.5 years before he joined Applus+. Mr. Simon Shen has extensive experience also as former Applus+ Shanghai CDM Technical Manager.*

2.2 Document Review

The PD version 01 dated 21/10/2024, version 04 dated 24/11/2025 and the emission reduction calculations spreadsheet /2/, were assessed as part of the validation. Relevant documents were reviewed. A detailed documents reviewed are listed in Appendix II of the report.

2.3 Interviews

The key personnel interviewed are summarized in the table below:

Interviewed personnel	***Role***	***Organization***	***Subject***
Mr. Li Jinke	*Manager*	*Lichuan City BCCY New Energy Co., Ltd.*	*Status of the project (including PPs);* *Applicability of selected methodology;* *Baseline of the project and its updates;* *Emission factors and their updates;* *Monitoring plan;* *Stakeholder consultation process and its outcomes.* *The process and participation of the stakeholder consultation;* *The impact of the project activity;* *The complaint by local stakeholders and the implementation of the mitigation measures;* *Status of MSW landfill;* *Operation of MSW landfill.*
Ms. Meng Ting	*Accountant*	*Lichuan City BCCY New Energy Co., Ltd.*	
Mr. Tian Yongsheng	*Operator*	*Lichuan City BCCY New Energy Co., Ltd.*	
Mr. Wang Bo	*Staff*	*Lichuan landfill site*	
Mr. Zhao Lin	*Staff*	*Lichuan landfill site*	
Mr. Zhang Yongshu	*Villager*	*Yangliu Community*	
Mr. Xiao Jun	*Villager*	*Yangliu Community*	
Mr. Li Jianhua	*Villager*	*Yangliu Community*	
Ms. Zhang Yangyang	*Project Manager*	*Henan BCCY Environmental Energy Co., Ltd.*	*Data collection and ER calculation.*

2.4 Site Visits

The assessment team performed the on-site validation (Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, China) during 30/12/2024 - 31/12/2024. The interviewed personnel and objective are listed in above table. Following aspects have been checked through the site visit as below:

(a) Status of the project (including PPs);

(b) Applicability of selected methodology;

(c) Baseline of the project and its updates;

(d) Emission factors and their updates;

(e) Monitoring plan;

(f) Stakeholder consultation process and its outcomes;

(g) The process and participation of the stakeholder consultation;

(h) The impact of the project activity;

(i) The complaint by local stakeholders and the implementation of the mitigation measures;

(j) Status of MSW landfill;

(k) Operation of MSW landfill.

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the Project proponents, or if the evidence provided to prove conformity is insufficient;

b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the Project proponents;

c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;

d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the Project proponents.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for registration.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI

Technological Center S.A. (Applus+ Certification)'s positive conclusion on the PD. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation, the concerns raised, and responses given are summarized below in the Appendix III.

The PD version 04 dated 24/11/2025 serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation findings	*No. of CL*	*No. of CAR*	*No. of FAR*
Project Description and monitoring report	*00*	*00*	*00*
Description of project activity	*00*	*00*	*00*
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	*01*	*00*	*00*
- Deviation from methodology	*00*	*00*	*00*
- Clarification on applicability of methodology, tool and/or standardized baseline	*00*	*00*	*00*
- Demonstration of additionality	*00*	*00*	*00*
- Emission reductions	*00*	*00*	*00*
- Monitoring plan	*00*	*00*	*00*
- Stakeholders consultation process	*00*	*00*	*00*
- Environmental Impacts	*00*	*00*	*00*
- Public comments	*00*	*00*	*00*
Others (please specify)	*00*	*00*	*00*
Total	*01*	*00*	*00*

2.5.1 Forward Action Requests

No FAR was raised during the validation process.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity is a landfill gas recovery and power generation project of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2 MW consisting of 2 sets of 1,000 kW generators.

The project uses LFG from Lichuan landfill site for power generation. During the 10-year fixed crediting period, the power generated by the project is expected to be 75,670 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 456,696 tCO_2e during the 10-year fixed crediting period.

The project activity is located at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. The project activity has been developed by Lichuan City BCCY New Energy Co., Ltd. The geographic coordinates of the project activity are longitude of 108°58′53.205″ E and latitude of 30°18′1.403″ N which is confirmed by site visit.

The Project activity has been approved by Chinese government by checking the Project approval /16/ and Environmental Impact Assessment (EIA) approval /18/. The project started construction on 20/04/2024 confirmed by checking construction order /20/ and commissioned on 05/09/2024 by checking operation log /25/ and site visit.

Assessment team checked through on-site audit and confirms that the details of the project proponent is as below:

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Minister of carbon emission department
Address	10/F, Kailin Building, 99 Ruyi West Road, No.22, Zhengzhou, Henan Province, P. R. China
Telephone	+86 371 56737901
Email	lwang@bccynewpower.com

Assessment team checked through on-site audit and confirmed that the details of the other entity involved is as below:

Organization name	*Lichuan City BCCY New Energy Co., Ltd.*
Role in the project	*Project Operation Entity*
Contact person	*Yangyang Zhang*
Title	*Project Manager*
Address	*10/F, Kailin Building, 99 Ruyi West Road, No.22, Zhengzhou, Henan Province, P. R. China*
Telephone	*+86 371 65521780*
Email	*zhangyangyang@bccynewpower.com*

By checking cooperation agreement between Lichuan City Urban Management Law Enforcement Bureau and Henan BCCY Environmental Energy Co., Ltd. /26/ which is the mother company of Lichuan City BCCY New Energy Co., Ltd. regarding to the utilization of landfill gas, the cooperation period for LFG generation will last after 04/09/2034. By checking land certificate /39/, the landfill site solely belonged to Lichuan City Environmental Sanitation Service Center which is the owner of the landfill site last for 50 years. And according to venue lease contract /43/ signed by Lichuan City Environmental Sanitation Service Center and project proponent, the project has the right to use the land located at the landfill site. Also, according to the cooperation agreement signed by the project proponent and Lichuan City Urban Management Law Enforcement Bureau, the project proponent has the right to use the landfill gas generated by the landfill site until the shutdown of the power generation project not the landfill site. Therefore, it is confirmed that even after 2026 which is the time the landfill plan to shut down, the project proponent still has the right to use the landfill gas and landfill site until the shutdown of the power generation project.Overall, it is confirmed that the PD is accurate, complete, and provides an understanding of the nature of the project.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	*The validation of this project activity was conducted by LGAI Technological Center, S.A. (Applus+ Certification).*
Sectoral scope	*By checking PD and applied methodology, it is confirmed that the project falls under sectoral scope 01 and 13.*
AFOLU project category, if applicable	*Not applicable since the project activity was not an AFOLU project.*

Project activity type	*By checking PD and applied methodology, it is confirmed that the project activity type of the project belongs to Energy industries (renewable-/non-renewable sources) (scope 01) and Waste handling and disposal (scope 13). Also it is confirmed that the project activity is not a grouped project activity and has been designed as the single location or installation only.*
General eligibility of the project to participate in the VCS Program	*By checking VCS Standard Version 4.7, PD and site visit witness, it is confirmed that the project is excluded under Table 2.1 of the VCS Standard as a landfill gas power generation project.* *The project fully met the requirements related to pipeline listing deadline, the opening meeting with the VVB, and the validation deadline by checking information on the VCS website.* *By checking VCS website, it is confirmed that the applied methodology for the project is eligible under the VCS Program as small-scale methodology.* *By checking Project approval and site visit, it is confirmed that the project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.* *No other relevant eligibility information detected during the validation process.*
AFOLU project eligibility, if applicable	*Not applicable since the project activity was not an AFOLU project.*
Transfer project eligibility, if applicable	*Not applicable since the project activity is not a Transfer project.*
Project design	*Not appliable since the project is not a grouped project activity.*
Project ownership	*Lichuan City BCCY New Energy Co., Ltd. is the project owner of project activity and its mother company Henan BCCY Environmental Energy Co., Ltd. is project proponent (PP) of project activity, and they have the legal right to control and operate the project activity.* *The project ownership has been checked by the assessment team and demonstrated through checking business license /14/, Project Approval and EIA Approval.*

Project start date	*By checking operation log and via onsite interview, it was verified the project started commercial operation on 05/09/2024, which is the start date of the project activity.*
Project crediting period	*The project activity adopts 10-year fixed crediting period.* *The assessment team confirms that the crediting period dates for the project is as below:* *Crediting Period Start date: 05/09/2024* *Crediting Period End date: 04/09/2034*
Project scale	*Assessment team confirms that the project is small-scale project under CDM scheme. The total installed capacity of the project activity is 2 MW less than 15 MW for Type I component. Although for certain year, the total emission reduction is higher than 60,000 tons (e.g. 2026), but the project results in aggregate emission reduction of less than 60kt CO_2 equivalent annually from all Type III components (LFG utilization part). Thus, small scale methodology is still applicable for the project.* *As the estimated annual average GHG emission reductions or removal per year is 45,669 tCO_2e which is less than 300,000 tonnes of CO_2e per year, thus the project falls in the category of Project under VCS Standard.*
Likelihood of achieving estimated GHG emission reduction or removals	*As the estimated annual average GHG emission reductions or removal per year is 45,669 tCO_2e, which is less than 300,000 tCO_2e per year, thus the project falls in the category of Project.*

Year	***Estimated GHG emission reductions or removals (tCO_2e)***
05/09/2024 – 31/12/2024	*17,847*
2025	*59,323*
2026	*63,430*
2027	*56,543*
2028	*50,640*
2029	*45,566*
2030	*41,188*
2031	*37,398*
2032	*34,104*

	2033 | *31,229* *01/01/2034 – 04/09/2034* | *19,428* *Total estimated ERs* | *456,696* *Total number of crediting years* | *10* *Average annual ERs* | *45,669* *The above estimated emission reduction is confirmed by assessment team via emission reduction calculation spreadsheet. The calculation is conservative and this acceptable to the assessment team. The likelihood of achieving estimated GHG emission reduction is high.*
Technologies and measures implemented by the project activity or activities	*The project activity is a landfill gas recovery and power generation project of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2 MW consisting of 2 sets of 1,000 kW generators.* *The project consists in LFG collection, transmission and treatment system, with subsequent electricity generation and grid connection system.* ***LFG collection system*** *LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.* ***LFG treatment system*** *Prior to electricity generation, LFG is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.* ***Electricity generation and grid connection system*** *2 gas engines of 1,000 kW rated power each are fed with the LFG and generate electricity, which is then exported to the grid.*

The technical specifications are mentioned as below and the same were checked during on-site visit and checking technical specification of equipment /28/:

Main Equipment	***Parameter***	***Value***
Generator set	*Type*	*1000GF-NK*
	Manufacturer	*JINAN DIESEL ENGINE CO., LTD*
	Rated capacity	*1,000 kW*
	Rated rotation speed	*1,000 r/min*
	Rated frequency	*0.8 (lagging)*
	Lifetime	*15 years*
LFG collection system	*Type*	*An integrated system consisting of gas collection wells and pipelines*
LFG treatment system	*Type*	*LFG-1500*
	Manufacturer	*Nanjing Carbon Ring Biomass Energy Co., Ltd.*
	Handle gas volume	*1,500 Nm^3/h*
Rotary air compressor	*Type*	*3L42WC*
	Number	*2*
	Manufacturer	*NANTONG HENGRONG BLOWER AND PUMP CO., LTD*
	Flow volume	*17.18 m^3/min*
	Motor power	*22 kW*
	Speed	*1,450 r/min*

	Type	*0L52WC*
	Number	*1*
	Manufacturer	*NANTONG JULONG FLUID MACHINERY CO., LTD*
	Flow volume	*32.8 m³/min*
	Motor power	*37 kW*
	Speed	*1,450 r/min*

By site visit interview, it is confirmed that the project does not include any flare system. The project uses recovered landfill gas to generate electricity with a destruction efficiency of 100 percent in the gas generator. During the operation period, the LFG flow can be adjusted based on LFG demand of the project. Even in case all the gas generators have to stop operation for emergency and the landfill site is able to hold LFG. The solenoid valve on the intake pipeline at the front end of each gas generator will also be automatically activated. The LFG will be temporarily stored in the landfill site and pipelines and impossible pass into the atmosphere. When the engines are scheduled to shut down, the LFG collection system reduces the gas in advance. There was no LFG collected and left in the entire project system during the planned shutdown. At this point in time, LFG did not enter through the roots blower pumps, LFG was stored in the pipeline between the landfill and the pre-treatment facility, the landfill was equipped with gas wells that have the capacity to hold the amount of LFG generated. At the beginning of the design process, the design took into account the length of the pipeline and the safety features, and the pipeline has the capacity to store the LFG. Safety has been taken into account during the design and commissioning stages of the project and the design is compliant. When the unit is shut down in an emergency, intake pressure is increased and the roots blower pumps will reduce its frequency and extract less LFG from the landfill site. It is a type of interlocking control, ensuring that the pressure in the unit is balanced and that a very small portion of the LFG stored in the pipework can be accommodated. At the same time, the

	maintenance personnel will handle the malfunction as soon as possible. So, the LFG won't pass into the atmosphere. After the engines return to normal, the LFG collection system will adjust to increase the intake volume.
Implementation schedule of the project activity or activities	*The Project activity has been approved by China's government by checking Project Approval and Environmental Impact Assessment (EIA) approval. The project started construction in 20/04/2024 confirmed by construction order and commissioned on 05/09/2024 by checking operation log and on-site visit interview.*
Project location	*The project activity is located at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, China. The project activity has been developed by Lichuan City BCCY New Energy Co., Ltd. The geographic coordinates of the project activity are longitude of 108°58′53.205″ E and latitude of 30°18′1.403″ N which is confirmed through checking online GPS map during site visit.*
Conditions prior to project initiation	*Before the implementation of the project activity, the electricity generated by the project would be supplied by CCPG in the baseline scenario and the LFG combusted would be vented to atmosphere without utilization.* *The same has been confirmed by checking FSR, Project approval and site visit.*
Project compliance with applicable laws, statutes and other regulatory frameworks	*Assessment team confirms that the Project has been approved by P. R. China's government by checking the Project approval and Environmental Impact Assessment (EIA) approval.* *By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.*
Double counting and participation under other GHG programs	***Projects registered (or seeking registration) under other GHG program(s)*** *The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. The declaration /29/ for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries*

	to confirm the same. The details of the registries checked are as follows: *- http://www.irecstandard.org/* *- http://cdm.unfccc.int/* *- http://www.goldstandard.org/* *- https://ccer.cets.org.cn/* *- http://verra.org/* ***Rejection by other GHG programs*** *The Project is not rejected by other GHG programs. A declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:* *- http://www.irecstandard.org/* *- http://cdm.unfccc.int/* *- http://www.goldstandard.org/* *- https://ccer.cets.org.cn/* *- http://verra.org/* *It is confirmed that the project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. Applus+ Certification checked the REC Mechanism database of P. R. China and found that the project activity is not accredited / registered under REC mechanism. Furthermore, although China has re-launch domestic GHG program as CCER, based on current regulation, the project is not eligible under CCER. Further, declaration for the same is checked and found correct by the assessment team.* *The project has provided all information required on whether is registered or seeking registration under any other GHG programs as the project only apply under VCS Standard.* *The Project is not rejected by other GHG programs as the project only apply under VCS Standard.*

No double claiming with emissions trading programs or binding emission limits	*The assessment team confirms that the net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. The assessment team checked the REC Mechanism database of China and Chinese Emission Trading System (Chinese ETS) and found that the project activity is not accredited / registered under REC mechanism or Chinese ETS. Thus, the assessment team concluded that the project activity is not involved on other Emissions trading programs and other binding limits.* *Furthermore, as per "Notice on Key Work Related to the Management of Greenhouse Gas Emission Reports for Enterprises in 2022" /30/ issued by Ministry of Ecology and Environment of P. R. China, China has a national emissions trading scheme covering thermal power generation industry. However, power generation using landfill gas is excluded and thus no emission cap was enforced for the project proponent.*
No double claiming with other forms of environmental credit	*The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program. Renewable energy certificates are available for trading in the host country. By checking REC Mechanism database of China /31/ as well as I-REC website /32/, it was verified only covering wind, solar, hydro and biomass power was covered. Therefore, it was verified the project activity is not included in REC Mechanism database of China.* *In conclusion, the assessment team confirmed the project activity is not involved in other forms of environmental credit.*
Supply chain (Scope 3) emissions double claiming	*It is confirmed that the project is to capture and utilize the landfill gas which otherwise will be released to atmosphere, and it will not impact the emissions of goods and services in a supply chain, i.e. Scope 3 emissions.*
Sustainable development contributions	*The project activity would contribute sustainable development in the region in following aspects confirmed by on-site audit:* ***SDG 7 Affordable and Clean Energy***

Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Central China Power Grid and further help to achieve the national action to promote renewable energy development;

SDG 8 Decent works and Economic Growth

The project will provide new jobs opportunities and increase tax revenue, which will have a positive effect on the local economy;

SDG 13 Climate Action

The project will recover and destroy landfill gas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030.

Sustainable development priorities are illustrated in the People's Republic of China National Report on Sustainable Development issued on 01/06/2012. Sustainable development priorities of the Report related to the project are promoting new energy development, strengthening environmental pollution control and responding to climate change. Methane is an ideal clean fuel. Each cubic meter of methane contains about 360,000 kJ calorific value, LFG recovery and utilization will supplement the energy supply of the grid and contributes to the new energy development in China. Odour from landfill negatively impacts on residents around the landfill. Implementation of the project will reduce odours through LFG collection and will thus mitigate the impact of landfill odour on people's daily lives. Landfill gas contains nearly 50% of methane and methane is a greenhouse gas some 28 times more potent than carbon dioxide. LFG recovery and utilization is contribution to mitigating global warming.

SDG Target	***SDG Indicator***	***Project activity contribution***
13.0	*Tonnes of greenhouse gas emissions*	*The project avoided anthropogenic emissions of*

		avoided or removed	*greenhouse gases (GHG) of 456,696 tCO_2e during the crediting period (05/09/2024 to 04/09/2034).*
	8.5	*Total number of job opportunities*	*The project activity employs 8 permanent and full-time employees through its operational company Lichuan City BCCY New Energy Co., Ltd. during the crediting period (05/09/2024 to 04/09/2034).*
	7.2	*MWh of renewable energy generated*	*Total 75,670 MWh of renewable energy generated during the crediting period (05/09/2024 to 04/09/2034).*
Additional information relevant to the project	***Leakage management for AFOLU projects*** *Not applicable to the project activity.* ***Commercially sensitive information*** *No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation.* ***Any additional relevant information***		

	As confirmed with PP, the assessment team is able to confirm that there are no additional relevant information that may have a bearing on the eligibility of the project, the reductions or removals, or the quantification of the project's reductions or removals.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	*By checking PD and site visit, the stakeholder involved in the project has been correctly identified and in line with the actual situation as below:* *The project is located in Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. local authority, employee of Lichuan City Urban Management Law Enforcement Bureau (landfill site), employee of project and local residents are therefore considered as stakeholders. Surely the project proponent is also considered as stakeholder.* *The assessment team considers the stakeholder identification of the project activity as reasonable.*
Legal or customary tenure/access rights	*The project is located in Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China.* *By checking Cooperation agreement signed by the project owner and Lichuan City Urban Management Law Enforcement Bureau, it is confirmed that the project owner has the legal right to use the LFG generated from Lichuan landfill site for power generation.* *By checking venue lease contract signed with Lichuan City Environmental Sanitation Service Center (the operation entity of Lichuan landfill site), the project has the right to use the land located at the landfill site.*

Stakeholder diversity and changes over time	*As stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time.*
Expected changes in well-being	*By site visit and checking Project approval, it is confirmed that expected changes in well-being as below:* *1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected;* *2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 75,670 MWh, which could replace the equivalent electricity from fossil fuel based grid.*
Location of stakeholders	*The project is located in Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China, local authority, employee of Lichuan City Urban Management Law Enforcement Bureau (landfill site), employee of project and local residents are therefore considered as stakeholders.* *The assessment team considers the stakeholder identification of the project activity as reasonable.*
Location of resources	*The location of resources is at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China.*

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	*By checking PD, Stakeholder Meeting Notice /23/, Stakeholder Meeting Minutes /21/ and interviewing stakeholders and PP during site visit, it was verified a stakeholder meeting was held on 19/04/2024 to collect stakeholders' comments for the project activity. Via interviewing stakeholders and PP during site visit, it was verified before the stakeholder consultation meeting, the project information with contact information were published on the bulletins at and nearby the project site on 18/03/2024 to invite Local*

	stakeholders. *By checking questionnaires /22/, it was verified a survey was carried out to collect comments from local residents through distributing and collecting responses to a easy-designed questionnaire. PP considered the differences and interactions between the stakeholder groups, the stakeholders involved in the questionnaire survey were thereby local residents with different sex, age groups, education background and occupation, which are within different social, economic, and cultural diversity. The questionnaire includes a brief summary of the project activity, including the project design and other relevant key information. In total 13 out of 13 questionnaires were returned with a 100% response rate.*
Consultation outcome	*By checking questionnaires, meeting minute and site visit, it is confirmed that during the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project owner will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures.*
Ongoing communication	*By checking grievance expression book and site visit, it is confirmed that except for introducing the project information, grievance mechanism was also explained during the local stakeholder consultation meeting. A grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project Proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Proponent will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible. Also, the project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting*

	every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness. *Therefore, it is able to confirm the ongoing communication process with the local stakeholders and grievance handling procedure is in place with effectiveness.*
Stakeholder input	*By checking the questionnaires, meeting minutes and site visit, Applus+ Certification confirm that the local stakeholder has no major negative comments for the construction of the project activity, and minor negative comments have been considered with mitigation measurements.*

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government.* *By checking Cooperation agreement signed by the project owner and Lichuan City Urban Management Law Enforcement Bureau, it is confirmed that the project owner has the legal right to use the LFG generated from Lichuan landfill site for power generation.* *By checking venue lease contract signed with Lichuan City Environmental Sanitation Service Center (the operation entity of Lichuan landfill site), the project has the right to use the land located at the landfill site.* *By checking questionnaires, grievance expression book, meeting minutes and site visit, it is confirmed that local stakeholders held full right for FPIC.* *Therefore, it was verified the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.*
Outcome of FPIC discussion	*As mentioned above, the project has been approved by Chinese government and local management institute. And local stakeholders have no major negative comments for the construction and operation of the project.*

	The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	*By checking grievance expression book /24/ and site visit, it is confirmed that a grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Also, the project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness.* *Therefore, it is able to confirm the ongoing communication process with the local stakeholders and grievance handling procedure is in place with effectiveness.*
Grievance redress procedure	*By checking grievance expression book and site visit, it is confirmed that a grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Also, the project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness.* *Therefore, it is able to confirm the ongoing communication process with the local stakeholders and grievance handling procedure is in place with effectiveness.*

3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
The assessment team noted that the project activity was open for public comment from 19/11/2024 to 19/12/2024. The detail was checked by the assessment team in the following web platform: https://registry.verra.org/app/projectDetail/VCS/5313. *During this period, 4 public comments were received. Please refer to the table below for each.*	*In the PD, response and clarification have been made to each comment. No further action was made. Please refer to details as below.*	*By checking response and clarification made by the project proponent in the PD, the assessment team consider the comments have been well responded and in line with related regulation and common practice. Also, all issues have been covered in the assessment process. No further action was required. See below the justification for each comment below.*
[PG 50] Noting that the PD has opted for the Simple OM calculation and has chosen to employ the exante option for its operation margin calculation, i.e., the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. a. It will be imperative for Verra team to request for a rationale for the decision to have opt for the ex-ante option, given that China's renewable energy mix in its grid is expected to increase overtime (at least, substantial change is expected within the project's 10-year crediting lifetime). b. It appears to me that ex-post	*Firstly, in China, the data for OM calculations is not available. Therefore, we can only use the latest available OM data published by DNA (in China, the DNA is Ministry of Ecology and Environment). In the OM calculation file released by DNA on 08/06/2024, DNA clearly stated in accordance with the requirements of Tool to calculate the emission factor for an electricity system (version 07.0) that the OM emission factors for the Central China Power Grid in 2023 were calculated using the simple OM method, and were ex-ante calculated.*	*According to Tool 07: Tool to calculate the emission factor for an electricity system (version 07.0), project participants may delineate the project electricity system using any of the following options: (a) Option 1. A delineation of the project electricity system and connected electricity systems published by the DNA or the group of the DNAs of the host country(ies), In case a delineation is provided by a group of DNAs, the same delineation should be used by all the project participants applying the tool in these countries; (b) Option 2. A delineation of the project electricity system defined by*

option would provide a more accurate accounting methodology of the GHG emissions that have been avoided through the displacement of the grid electricity.	*Secondly, by combining the available OM emission factor data for CCPG over the past three years (2021: 0.7938 tCO_2/MWh, 2022: 0.8075 tCO_2/MWh, 2023: 0.8771 tCO_2/MWh), it was found that the values were gradually increasing. Based on this trend, the OM emission factor of CCPG is likely to increase after 2023. If applied ex-post option, the calculation of the reduction amount will also increase. Therefore, in order to conservative calculation, this project adopts ex-ante option.* *In conclusion, it is reasonable for this project to adopt the Simple OM and ex-ante calculation method, and it also meets the requirements of the methodology and the tools.*	*the dispatch area of the dispatch centre responsible for scheduling and dispatching electricity generated by the project activity. Where the dispatch area is controlled by more than one dispatch centre, i.e. layered dispatch area, the higher level area shall be used as a delineation of the project electricity system (e.g. where regional dispatch centres are required to comply with dispatch orders of the national dispatch centre then area controlled by the national dispatch centre shall be used); (c) Option 3. A delineation of the project electricity system defined by more than one independent dispatch areas, e.g. multi-national power pools.* *Therefore, it is the project proponent's choice to choose one of above (3) options as the tool provides such choices, and rational for making such choice is not required by the tool. However, justification has been made by project proponent as below:* *(1) Calculation data for OM is not available for public; all OM calculation was calculated by China DNA in accordance with the*

		requirements of Tool to calculate the emission factor for an electricity system (version 07.0) and then public. The latest version available has been used for ER calculation. (2) By comparing the OM data published by China DNA in recent 3 years, it is found the data increased yearly. It is even more conservative to adopt ex-ante option for OM calculation. In conclusion, the data adopted for OM for Emission factor is accurate and met all requirement of methodology and tools, also this is the common practice for all projects in China.
[PG 52] Noting that there is electricity exchange between Central China Power Grid and the Southern Power Grid, and that the emission factors are released nationally by the Chinese authorities. Based on my interpretation, the calculation of the adjusted grid emission factor used based on a weightage average of the annual power supply of the grid across all 7 power grids in China was considered, but not adopted – since the eventual OM grid emission factor used was 0.8771 tCO_2e/MWh,	*Firstly, Ministry of Ecology and Environment of the People's Republic of China has published a delineation of the project electricity system and connected electricity systems, so the project adopt the delineation of project electricity system and connected electricity system published by Ministry of Ecology and Environment of the People's Republic of China. The power generated by the project displaces the equivalent electricity*	*By checking 2023 Baseline Emission Factors for Regional Power Grids in China published by China DNA, it is confirmed that the response by the project proponent is accurate, the commentor has mere understanding for Emission Factor in China.* *All Emission Factor data are provided by China DNA which is Ministry of Ecology and Environment of the People's Republic of China calculated in accordance with the requirements of Tool to*

which is the OM grid emission factor for Central China (link). a. I would like to respectfully challenge whether a weightage average of Central China Power Grid and Southern Power Grid should be taken into consideration, in which the PD can consider using the lower of (average emission factors of Central China Grid 2023 – 2021 since Central China National Grid is the injection point of the project) and (weighted average emission factor across both Central China and Southern Power Grids in China 2023 – 2021). Rationale being, from a more conservative perspective, doing a lower of both methodologies will ensure that the emissions avoided is not overinflated.	*generated by Central China Power Grid.* *Secondly, Central China Power Grid is a larger regional grid, which consists of 4 sub-grids: Henan Province, Hubei Province, Hunan Province and Jiangxi Province. In addition, there is net imported power to Central China Power Grid from the North China Power Grid, Southwest China Power Grid and Northwest China Power Grid. According to the "Tool to calculate the emission factor for an electricity system", the project chose option (b) to calculate the CO_2 emission factor for net electricity imports from the Central China Power Grid.* *Thirdly, according to data released by DNA, the OM factor of CCPG was the weighted average of the simple OM emission factors or adjusted simple OM emission factors for each of the last three years (2019-2021) for which data was available, with the power grid supply as the weight (included net imported power from the North China Power Grid, Southwest China Power Grid and Northwest China Power Grid). Similarly, the OM factor of Southern China Power Grid was also the*	*calculate the emission factor for an electricity system (version 07.0).* *This is the common practices for project in China. The data of EF is clearly provided by government.*

	weighted average of the simple OM emission factors or adjusted simple OM emission factors for each of the last three years (2019-2021) for which data was available, with the power grid supply as the weight (included net imported power from the Central China Power Grid and East China Power Grid). If the project considered a weightage average of Central China Power Grid and Southern Power Grid, the OM factor will be double-counted. Therefore, the project chose OM factor of CCPG to calculate is reasonable and in line with Tool 07.	
Noting that for most other data/parameters, feasibility study report (FSR) is used as the basis of justification. However, the "Energy Conversion Efficiency of the project equipment" used the 40%, which is a default factor. I would like to clarify the rationale of the Project Developer of using so – given that the FSR should have utilised or consulted industry players on the efficiency range.	*According to technical agreement of the generator set, the Energy Conversion Efficiency is 33%-36%. The project used 40% is more conservative. The reasons are as follows. According to AMS-III.G. (Version 10.0), the project can calculate $FC_{H4,PJ,y}$ using equation (6), in this equation, the Energy Conversion Efficiency (EEy) appears as the denominator. If the project uses 33%-36% to calculate, the result is larger that using 40%. Therefore, the project uses 40% is more conservative. And it is also in line with the*	*According to AMS-III.G, the methodology provides 2 options for determination of this value:* *(1) Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation.* *(2) Default efficiency of 40 per cent.* *Based on technical specification of equipment of*

	requirements of AMS.III.G. (Version 10.0).	*the generator, the Energy Conversion Efficiency is around 33~36%. Therefore, it is more conservative to apply 40% as the default value.*
Finally, in the entire PDD, there was no mention of treatment of transmission/distribution losses, which should result in 5 – 15% of reduction of the eventual consumption of total renewable electricity generated from the plant, hence it might be helpful for the PD to reach out to the provincial grid authority to have a estimative range of transmission losses, so that this can also be accounted for in the estimated tCO_2e avoided via energy source replacement.	*As mentioned in the methodology ACM0001 (version 19.0) applicable to large-scale project, when calculating $BE_{EC,y}$, the equation (2) in Tool 5, namely $BE_{EC,y}=EC_{BL,k,y}*EF_{EF,k,y}*(1+TDL_{j,y})$ needs to be used. When calculating $PE_{EC,y}$, the equation (1) in Tool 5 needs to be used, that is, $PE_{EC,y}=EC_{PJ,k,y}*EF_{EL,k,y}*(1+TDL_{j,y})$. However, this project is a small-scale project. Methodologies AMS-I.D: Grid connected renewable electricity generation (version 18.0) and AMS-III.G: Landfill methane recovery (version 10.0) are applied in this project. According to AMS-I.D (version 18.0), when calculating Baseline emissions include only CO_2 emissions from electricity generation in powerplants that are displaced due to the project activity, equation (1) is used, that is $BE_y=EG_{PJ,y}*EF_{grid,y}$. According to AMS-I.D (version 18.0) and AMS-III.G (version 10.0), the project emissions for this project are zero (the specific reasons have been described in Section 4.2 of this*	*As the project applied small-scale methodology, when calculating baseline emission, the TDL was not considered which is conservative. For the calculation of project emission as explained in the below report was considered as zero. Therefore, no TDL is involved in this project.*

	document and will not be repeated here) PE_y*=0. In summary, this project calculates the emission reduction based on the equations of AMS-I.D (version 18.0) and AMS-III.G (version 10.0). The equations do not involve the parameters of transmission and distribution losses. Therefore, the calculations for this project are reasonable and comply with the methodological requirements. Therefore, the project is not involved in transmission/distribution losses.*	

3.2.2 Risks to Local Stakeholders and the Environment

3.2.2.1 Management Experience

By checking Business License, it was verified the PP was established on 10/01/2024 with main business scope containing "power generation". Therefore, the assessment team deemed PP has expertise or experience in implementing similar project activities and engaging communities.

By checking EIA Approval and Project Approval issued by government, it is confirmed the project has been approved by Chinese government which means the government recognize the experience of PP for project construction and operation. Moreover, the project owner of the project has developed several LFG projects and therefore has extensive development and management experience.

3.2.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	*By checking EIA Approval and Project Approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project meets all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.*

	On the other hand, the project would contribute following changes in well being according to project approval: *1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected;* *2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 75,670 MWh, which could replace the equivalent electricity from fossil fuel based grid.* *Therefore, it was verified no natural and human-induced risks has been identified to stakeholders' wellbeing.*
Risks to stakeholder participation	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.* *Also, during the site visit, no risks have been identified to stakeholder participation.* *Therefore, it was verified no risks have been identified to stakeholder participation.*
Working conditions	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.* *Also, during the site visit, it is confirmed that the staff of the project is working under indoor environment and safety protection equipment have been provided in case for on-site monitoring and device maintenance. No risks identified for working conditions.*
Safety of women and girls	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet*

	all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. Moreover, China has published "Women's Rights Protection Law" /33/ to provide support and protection to women and girls. *In order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations" /33/, prohibiting employers from recruiting minors under the age of 16. Also, China published "Women's Rights Protection Law". Therefore, the Project will not expose women and girls to further risks or hazards. By checking HR Records and Staff Roaster /34/, it was verified no girls have been recruited, also equal payment would be made to women and men in case.* *Also, during the site visit, it is confirmed that no risks has been identified to safety of women and girls.*
Safety of minority and marginalized groups, including children	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of minor protection.* *The country has strict prohibition for child labour. Thus, project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.* *Besides, in order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations", prohibiting employers from recruiting minors under the age of 16. By checking HR Records and Staff Roaster and site interview, the assessment team was able to confirm that no Minors under the age of 16 were employed.* *Also, during the site visit, it is confirmed that no minors have been hired by the project owner. No child labor detected for the project. No risks identified for safety of minority and*

	marginalized groups, including children as not applicable for the project.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials, and chemical pesticides and fertilizers)	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. During construction phase, the raise dust, noise, wastewater and solid wastes caused by project construction will be treated according to the measures in EIA report /18/, and there will be no significant impact on the environment.* *During the operation period, all the mitigation measures proposed by EIA report will be implemented and the following key aspects will be addressed:* ***Waste water*** *The wastewater of this project is mainly biogas condensate water and domestic sewage. Biogas condensate is collected by sewage tank, and domestic sewage is pretreated by septic tank. These wastewater then enters the landfill leachate treatment station, after treatment to meet the "Domestic Waste Landfill Pollution Control Standard" (GB16889-2008) and then into the municipal sewage network.* ***Waste gas*** *The air pollutants produced by this project are mainly methane combustion waste gas and odor gas produced in the process of landfill gas collection and transport. After the combustion exhaust gas is treated to meet the standard, it is discharged through 15m exhaust cylinder. The emission concentration is in line with the emission limit of "Air Pollutant Emission Standard for Thermal Power Plant" (GB13223-2011). The odor gas is transported by closed pipeline, and the emission is small, which meets the secondary standard of "Odor Pollutant Emission Standard" (GB14554-93).* ***Noise pollution*** *Noise sources include equipment noise generated during the operation of drilling RIGS, air compressors, generator sets*

	and various pumps. Through the selection of low noise equipment, reasonable layout, sound insulation, shock absorption and other measures, the factory boundary noise can meet the "Industrial Enterprise Factory Boundary Environmental Noise Emission Standard" (GB12348-2008) class 2 standards. ***Solid waste*** *Solid waste includes domestic waste, general solid waste and hazardous waste. The particulate matter and waste filter element produced in the process of biogas purification belong to general solid waste. The particulate matter is collected and sent to the landfill for treatment, and the waste filter element is collected and returned to the manufacturer for recycling. Waste oil and waste catalyst are classified as hazardous waste. They are collected separately and temporarily stored in hazardous waste storage before being disposed of by qualified units. Domestic waste is collected in garbage bins and sent to landfill for disposal.*

3.2.3 Respect for Human Rights and Equity

3.2.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" /33/ and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no discrimination and sexual harassment detected.* *The project does not create any direct or indirect impacts on gender equality and/or the situation of women:* *a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16*

	were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls. *b. The Project will not restrict women's rights or access regarding natural resources.* *c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.*
Sexual harassment	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no discrimination and sexual harassment detected.* *The project does not create any direct or indirect impacts on gender equality and/or the situation of women:* *a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls.* *b. The Project will not restrict women's rights or access regarding natural resources.* *c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.* *Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.*
Equal pay for equal work	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.*

	By site visit, it is confirmed no discrimination and sexual harassment detected. *The project does not create any direct or indirect impacts on gender equality and/or the situation of women:* *a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls.* *b. The Project will not restrict women's rights or access regarding natural resources.* *c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.* *Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.*
Gender equity in labor and work	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no Gender equity in labor and work detected.* *The project, complying with the "Women's Rights Protection Law" and "Labor Contract Law" does not discriminate the local community on basis of gender or caste or religion and therefore equally serve to all.* *By checking HR Records and interviewing staff from the project owner, it was verified the Project design neither increase women's workload nor prevent them from engaging in other activities. The Project has equal opportunity for women and men to contribute both in volunteer and working positions. The project has HR policy and same is followed equally. The project owner has a specified HR policy that considers participation by both men and women.* *Besides, China also ratified relevant ILO core conventions on equality, namely Equal Remuneration Convention (Convention No 100).*

	Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.
Forced labor	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena.* *Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China.* *China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee.* *China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.* *By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.*
Child labor	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena.* *Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China.* *China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee.*

	China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview. *By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.*
Human trafficking	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena.* *Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China.* *China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee.* *China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.* *By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.*

3.2.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
None	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights*

	China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties--the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. *Therefore, it is confirmed that the project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in "Women's Rights Protection Law" to provide support and benefits to women.*

3.2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
None	*By checking EIA report approved by government, it is confirmed that no indigenous peoples present in or within the area of influence of the project, the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.* *Hence not applicable for the project.*

3.2.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
None	*By checking EIA report approved by government, it is confirmed that no indigenous peoples present in or within the area of influence of the project, the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.* *Hence not applicable for the project.*

3.2.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	*Not applicable as no property rights involved for the project.*
Summary of the benefit sharing plan	*Not applicable as no property rights involved for the project.*
Approval and dissemination of benefit sharing plan	*Not applicable as no property rights involved for the project.*

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	*By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.*
Soil degradation and soil erosion	*The Project's purpose is to supply electricity from the landfill gas. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.* *By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.*
Water consumption and stress	*The Project's purpose is to supply electricity from the landfill gas. Therefore, the impact of the Project on water consumption and stress is negligible.* *By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.*

3.2.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion

Species and habitat	*By checking EIA report approved by government, it is confirmed that the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.*
Areas needed for habitat connectivity	*By checking EIA report approved by government, it is confirmed that the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.*

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	*Not applicable*

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	*Not applicable*

	Evidence gathering activities, evidence checked, and assessment conclusion
Invasive species	*Not applicable*

3.2.4.3 Ecosystem conversion

Risks Identified	Evidence gathering activities and evidence checked
Ecosystem conversion	*Not applicable as the project is not a ARR, ALM, WRC or ACoGS project.*

3.3 Application of Methodology

3.3.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Title: Landfill methane recovery; Grid connected renewable electricity generation

Reference: *The project activity meets the eligibility criteria of small-scale project as the estimated annual average GHG emission reductions or removal per year is 45,669 tCO_2e which is less than 60,000 tonnes of CO_2e per year.*

Methodology: *Landfill methane recovery, AMS-III.G, version 10.0*

Grid connected renewable electricity generation, AMS-I.D, version 18.0

Type I and III: *Energy industries (renewable / non-renewable sources); Waste handling and disposal*

Sectoral scope(s): *01 and 13*

Tools referred with above methodology and applicable for project activity are:

- Tool 08: Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0

- Tool 04: Emissions from solid waste disposal sites, version 08.1

- Tool 07: Tool to calculate the emission factor for an electricity system, version 07.0

- Tool 32: Positive lists of technologies, version 04.0

3.3.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
AMS-III.G	*2. This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.*	*Applicable.* *By checking the FSR /15/, Project approval and on-site audit, it is confirmed that the Project consists in capturing landfill gas (which contains methane) from the Lichuan landfill site, which is used for disposal of residues from human activities.*
AMS-III.G	*3. Different options to utilise the recovered landfill gas as detailed in paragraph 4 for "AMS-III.H. : Methane recovery in wastewater treatment" (version 19.0) are*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the project utilizes the recovered LFG to generate electrical energy directly*

	eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard. *Paragraph 4 of AMS-III.H.:* *The recovery biogas from the above measures may also be utilised for the following applications instead of combustion/flaring:* *(a) Thermal or mechanical, electrical energy generation directly;* *(b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or* *(c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed:* *(i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints;* *(ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users;*	*which meet the requirement of 4 (a) of AMS-III.H. (version 19.0).*

	or(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users *(d) Hydrogen production;* *(e) Use as fuel in transportation applications after upgrading.*	
AMS-III.G	*4. Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.*	*Applicable.* *It is confirmed the annual ER of the project is less than 60,000 CO_2e from all type III components.*
AMS-III.G	*5. The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in the Lichuan landfill site.*
AMS-III.G	*6. This methodology is not applicable if the management of the solid waste disposal site SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the management of the solid waste disposal site SWDS) in the project activity is not deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity.*

	may include, for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS or changing the shape of the SWDS to increase methane production	

Methodology ID	Applicability condition	Assessment and conclusion
AMS-I.D	*4. This methodology is applicable to project activities that:* *(a) Install a Greenfield plant;* *(b) Involve a capacity addition in (an) existing plant (s);* *(c) Involve a retrofit of (an) existing plant (s);* *(d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or;* *Involve a replacement of (an) existing plant(s).*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the project is a greenfield project.*
AMS-I.D	*5. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:* *(a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;*	*Applicable.* *By checking FSR and site visit interview, it is confirmed this is not appliable as project is not a hydro power project.*

	(b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m²; *The project activity results in new reservoirs and the power density of the powerplant, as per definitions given in the project emissions section, is greater than 4W/m².*	
AMS-I.D	*6. If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel. the capacity of the entire unit shall not exceed the limit of 15 MW.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the total installed capacity of the project is 2 MW.*
AMS-I.D	*7. Combined heat and power (co-generation) systems are not eligible under this category.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the project does not involve in heat generation.*
AMS-I.D	*8. In the case of project activities that involve the capacity addition of renewable energy generation*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the*

	units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	*project does not involve in capacity addition.*
AMS-I.D	*9. In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the project does not involve in retrofit, rehabilitation or replacement.*
AMS-I.D	*10. In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.*	*Applicable.* *By checking FSR and site visit interview, it is confirmed the project is designed to install two engines to combusted LFG of Lichuan landfill site, which mainly contains 50% methane. The baseline for the electricity component: equivalent electricity generated by the project is supplied by Central China Power Grid (CCPG), which is dominated by fossil fuel-based power plants. The baseline for the electricity component is in accordance with procedure prescribed under 5.2 of methodology AMS-I.D*
AMS-I.D	*11. In case biomass Is sourced from dedicated plantations, the applicability*	*Not applicable.*

	criteria in the tool "Project emissions from cultivation of biomass "shall apply.	*By checking FSR and site visit interview, it is confirmed that the project does not involved in biomass.*

Methodology ID	Applicability condition	Assessment and conclusion
Tool 04 Emissions from solid waste disposal sites	*(a) Application A: The VCS project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "AMS-III.G: Landfill methane recovery). The methane is generated from waste disposed in the past, including prior to the start of the VCS project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (VCS-PD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e. g. measuring the amount of methane captured from the SWDS).*	*Applicable. The project collects the methane from the Lichuan landfill site and uses it to generate the electricity.*
Tool 04 Emissions from solid waste disposal sites	*(b) Application B: The VCS project activity avoids or involves the disposal of waste at SWDS. An example of this application of the tool is ACM0022, in which*	*Not applicable. The project collects the methane from the Lichuan landfill site and uses it to generate the electricity.*

	municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	

Methodology ID	Applicability condition	Assessment and conclusion
Tool 07 Tool to calculate the emission factor for an electricity system	*This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).*	*Applicable, this tool is applied to estimate the OM, BM and/or CM.*
Tool 07 Tool to calculate the emission factor for an electricity system	*Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an*	*Applicable, the emission factor for the project electricity system is calculated for grid power plants only.*

	option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the Project proponents, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
Tool 07 Tool to calculate the emission factor for an electricity system	*In case of VCS projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.*	*Not applicable, the project electricity system is located in China.*
Tool 07 Tool to calculate the emission factor for an electricity system	*Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.*	*In case there are biofuel involved, the value applied to the CO_2 emission factor of biofuels is zero.*

Methodology ID	Applicability condition	Assessment and conclusion
Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream	*Typical applications of this tool are methodologies where the flow and composition of residual or flared gases or exhaust gases are measured for the determination of baseline or project emissions.*	*Applicable. The project collects the methane from the Lichuan landfill site and uses it to generate the electricity. The flow of methane is measured for the determination of baseline emissions.*
Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream	*Methodologies where CO_2 is the particular and only gas of interest should continue to adopt material balances as the means of flow determination and may not adopt this tool as material balances are the cost effective way of monitoring flow of CO_2.*	*Not applicable.*
Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream	*The underlying methodology should specify:* *(a) The gaseous stream the tool should be applied to;* *(b) For which greenhouse gases the mass flow should be determined;* *(c) In which time intervals the flow of the gaseous stream should be measured; and* *(d) Situations where the simplification offered for calculating the molecular mass of the gaseous stream (equations (3) or (17)) is not valid (such as the gaseous stream is predominantly*	*Applicable.* *It is confirmed that the gaseous stream determined the baseline emissions is LFG which mainly contains fraction of CH_4, which corresponds to point (a) and (b).* *LFG combusted by the engines is measured continuously by a flow meter and recorded hourly, which corresponds to point (c).* *Point (d) is not applicable.*

	composed of a gas other than N_2).	

Methodology ID	Applicability condition	Assessment and conclusion
Tool 32 Positive lists of technologies	*The use of this methodological tool is not mandatory for the Project proponents of a VCS project activity for demonstrating their additionality.*	*Applicable.* *The project apply tool for demonstrating its additionality*
Tool 32 Positive lists of technologies	*This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.*	*Applicable.* *The project apply tool conjunction with large-scale methodology AMS-III.G which refers to this tool.*
Tool 32 Positive lists of technologies	*The positive lists as contained in section 5 of this tool are valid up to 10/03/2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the "Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools", there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.*	*Applicable.* *The project collects the LFG from the Lichuan landfill site and uses it to generate the electricity, with a total capacity of 2 MW, which conforms to 5.1.1 (a) "The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW".*

3.3.3 Project Boundary

As per AMS-III.G and AMS-I.D, the project boundary is the physical, geographical site of the landfill where the gas is captured and destroyed/used and the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

It is confirmed by the assessment team that the project boundary of the project activity includes the sites where the LFG is flared or used (e.g. power plant). The electricity used by the project activity is sourced from CCPG and electricity generated by LFG capturing of the project activity is connected to CCPG. Therefore, the project boundary includes the whole LFG related system (e.g. LFG collection, LFG pre-treatment system, LFG power generation system, etc.) and all grid-connected power plants in CCPG. According to China DNA, CCPG covers Henan Province, Hubei Province, Hunan Province and Jiangxi Province.

The sources and GHG gases involved for the Project activity are as below:

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO_2	No	CO_2 emissions from decomposition of organic waste are not accounted since the CO_2 is also released under the project activity
		CH_4	Yes	The major source of emissions in the baseline
		N_2O	No	N_2O emissions are small compared to CH_4 emissions from SWDS. This is conservative
	Emissions from electricity generation	CO_2	Yes	Major emission source
		CH_4	No	Excluded for simplification. This is conservative
		N_2O	No	Excluded for simplification. This is conservative
	Emissions from heat generation	CO_2	No	No heat generation is included in the project activity
		CH_4	No	No heat generation is included in the project activity
		N_2O	No	No heat generation is included in the project activity
	Emissions from the use of natural gas	CO_2	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity

Source		Gas	Included?	Justification/Explanation
		CH_4	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
		N_2O	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO_2	No	Excluded for no fossil fuel is used by the project activity
		CH_4	No	Excluded for no fossil fuel is used by the project activity
		N_2O	No	Excluded for no fossil fuel is used by the project activity
	Emissions from electricity consumption due to the project activity	CO_2	Yes	May be an important emission source
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from flaring	CO_2	No	Excluded for simplification
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO_2	No	No distribution of LFG using trucks and dedicated pipelines
		CH_4	No	No distribution of LFG using trucks and dedicated pipelines
		N_2O	No	No distribution of LFG using trucks and dedicated pipelines

3.3.4 Baseline Scenario

As per AMS-III.G, the baseline scenario for LFG is the situation where, in the absence of the project activity, biomass and other organic matter are left to decay within the project boundary, and methane is emitted to the atmosphere, possibly with capture of LFG and destruction through flaring to comply with regulations or contractual requirements. For electricity, if all or part of the *electricity generated by the project activity is exported to the grid, the baseline scenario for all or the part of the electricity exported to the grid is assumed to be electricity generation in existing and/or new grid-connected power plants.*

The project activity captures LFG to produce electricity and supplies to the grid. In the absence of the project, LFG from Lichuan landfill site is emitted directly into atmosphere, the equivalent amount of power would have been supplied by CCPG, which is fed mainly by fossil fuel fired plants.

Applus+ Certification confirms that the Project has been approved by Chinese government by checking the Project approval and Environmental Impact Assessment (EIA) approval. By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

The assessment team, therefore, concludes that the VCS PD conforms to the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and VCS via VCS Standard version 4.7.

The project activity captures LFG to produce electricity and supplies to the grid. According to GB50869-2013 has been implemented from 01/03/2014, Item 3.0.3, 4.0.2, 8.1.1, 10.1.1, 11.1.1, 11.6.1, 11.6.3, 11.6.4 and 15.0.5 are mandatory provisions and must be strictly implemented. But Item 11.1.3 is a voluntary provision, and it is a common practice in China that the LFG from landfill sites is vented to the atmosphere directly. By checking "2021 Urban Construction Statistical Yearbook" and "2021 Urban and Rural Construction Statistical Yearbook", it is confirmed that in the cities, there are a total of 1,407 harmless treatment plants, of which only 282 landfill sites have collected and used landfill gas or burned it or disposed the waste in other ways; in the counties, a total of 1,441 harmless treatment plants, of which only 61 landfill sites have collected and used landfill gas or burned it or disposed the waste in other ways. According to the "Report on Methane Emission Control and Resource Utilization" released by the Ministry of Ecology and Environment, by the end of 2020, there were a total of 1,871 sanitary landfills in cities and counties across the country, and 270 biogas power generation projects were installed and connected to the grid, accounting for only 14.4% of the total number of sanitary landfills. And the assessment team confirmed that no regulatory surplus required for project activity.

The assessment team confirmed the baseline of the project is in the absence of the project, LFG from Lichuan landfill site is emitted directly into atmosphere, the equivalent amount of power would have been supplied by CCPG, which is fed mainly by fossil fuel fired plants.

Applus+ Certification confirms that the Project has been approved by Chinese government by checking the Project approval and Environmental Impact Assessment (EIA) approval. By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

3.3.5 Additionality

Assessment for regulatory surplus

Based on the professional knowledge of the assessment team, it is able to confirm there are no regulatory surplus for LFG generation project in China.

Assessment of Prior consideration of carbon revenue

Following time lime of project has been assessed and confirmed to be accurate.

Events	***Time***	***Assessment***
Signed Cooperation Agreement	*26/12/2023*	*Confirmed by checking cooperation agreement*
Completion of FSR	*01/2024*	*Confirmed by checking FSR*
FSR approval	*27/02/2024*	*Confirmed by checking Project Approval*
Board resolution	*20/03/2024*	*Confirmed by checking Board resolution /02/*
Completion of Environmental Impact Assessment Report	*04/2024*	*Confirmed by checking EIA report*
EIA approval	*18/04/2024*	*Confirmed by checking EIA Approval*
VCS stakeholder consultation meeting	*19/04/2024*	*Confirmed by checking Stakeholder Meeting Minutes*
Construction date of the project	*20/04/2024*	*Confirmed by checking Construction Order*
The project was put into operation.	*05/09/2024*	*Confirmed by checking Operation log and site visit*

Additionality Assessment

As per the Methodological Tool "Positive lists of technologies", the project activities at new or existing landfills (greenfield or brownfield) are deemed automatically additional, if it is demonstrated that prior to the implementation of the project activities the landfill gas (LFG) was only vented and/or flared (in the case of brownfield projects) or would have been only vented and/or flared (in the case of greenfield projects) but not utilized for energy generation, and that under the project activities any of the following conditions are met:

(a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW;

(b) The LFG is used to generate heat for internal or external consumption;

(c) The LFG is flared.

The project activity uses LFG which is collected from Lichuan landfill site to generate electricity. Prior to the implementation of the project activity, the LFG from Lichuan landfill site was only vented, not utilized for energy generation. By interviewing local stakeholders, staff from local government, staff from Lichuan landfill site and checking website of local government, it is confirmed by the assessment team that only the project activity uses LFG from Lichuan landfill site to generate electricity. By checking Project Approval and EIA Approval, the total installed capacity of the project activity is 2 MW, which is below 10 MW. Therefore, the project activity is deemed automatically additional.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline emissions

<u>Baseline emissions are determined according to the following equation for LFG part:</u>

$BE_{y,1}=\eta_{PJ}\times BE_{CH4,SWDS,y}-(1-OX)\times F_{CH4,BL,y}\times GWP_{CH4}$

Where:

$BE_{y,1}$	=	*Baseline emissions associated with the SWDS in year y (tCO_2e/yr)*
η_{PJ}	=	*Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used*
$BE_{CH4,SWDS,y}$	=	*Methane emission potential of a solid waste disposal site (tCO_2)*
OX	=	*Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used*
$F_{CH4,BL,y}$	=	*Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (tCH_4)*
GWP_{CH4}	=	*Global Warming Potential for methane*

For the project, Methane emission potential of a solid waste disposal site ($BE_{CH4,SWDS,y}$) is calculated using first order decay (FOD) model as follows:

$$BE_{CH4,SWDS,y}=\varphi_y\times(1-f_{y)}\times GWP_{CH4}\times(1-OX)\times\frac{16}{12}\times F\times DOC_{f,y}\times MCF_y\times\sum_{x=1}^{y}\sum_{j}(W_{j,x}\times DOC_j\times e^{-k_j\times(y-x)}\times(1-e^{k_j})$$

Where:

$BE_{CH4,SWDS,y}$	=	*Baseline, project or leakage methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (tCO_2e/yr)*
x	=	*Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)*
y	=	*Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)*
$DOC_{f,y}$	=	*Fraction of degradable organic carbon (DOC) that can decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)*
$W_{j,x}$	=	*Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)*
φ_y	=	*Model correction factor to account for model uncertainties for year y*
f_y	=	*Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y*
GWP_{CH4}	=	*Global Warming Potential of methane*
OX	=	*Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)*
F	=	*Fraction of methane in the SWDS gas (volume fraction)*
MCF_y	=	*Methane correction factor*
DOC_j	=	*Fraction of degradable organic carbon in the waste type j (weight fraction)*
k_j	=	*Decay rate for the waste type j (1 / yr)*

j = *Type of residual waste or types of waste in the MSW*

The parameters required to apply the FOD model is determined as:

Parameter	Application A	Justification
ϕ_y	0.75	Baseline emissions: default values
OX	0.1	Default value
F	0.5	Default value
$DOC_{f,y}$	0.5	Default value
MCF_y	1.0	Default values (based on SWDS type)
k_j	Refer to Section 3.3.8	Default values (based on waste type)
$W_{j,x}$	Refer to Section 3.3.8	Estimated once, with the waste composition obtained from EIA.
DOC_j	Refer to Section 3.3.8	Default values (based on waste type)
f_y	0	Requirement from AMS-III.G and Tool 04

Determination of $F_{CH4,BL,y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as requirement in this section). The four cases in the following table are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Currently China has regulations in place to deal with the management of landfills and to encourage utilization of LFG. Those regulations are:

"Standard for Pollution Control on the Landfill Site of Municipal Solid Waste" (GB 16889-2008), which became effective in 2008, issued by the Environment Protection Administration.

"Technical Code for Municipal Solid Waste Sanitary Landfill" (GB 50869-2013), issued by the Ministry of Construction in 2013.

According to item 5.15 of GB16889-2008, if the designed landfill capacity is more than 2.5 million tons and the landfill thickness is more than 20m, methane utilization facilities or flare burning facilities shall be built to treat the landfill gas containing methane. For municipal solid waste landfills smaller than the above scale, technologies that can effectively reduce methane generation and emission shall be adopted or flare combustion facilities shall be used to treat methane containing landfill gas.

Item 11.1.1 of GB 50869-2013 stipulates that the landfill site must be equipped with effective landfill gas drainage facilities to prevent the natural accumulation and migration of landfill gas, causing fire and explosion. Item 11.1.3 stipulates that if the landfill does not have the conditions for landfill gas utilization, the flare method shall be adopted for combustion treatment, and the process that can effectively reduce the generation and emission of methane shall be adopted. The old landfills that are not safe and stable should be equipped with effective landfill gas drainage facilities. Among them, item 11.1.1 is mandatory and must be strictly implemented.

In fact, the LFG of Lichuan landfill site is emitted to atmosphere without LFG capture system prior the implementation of the project. Therefore, Case 2 listed in the table above is applicable for the project.

The requirements above don't specify any amount or percentage of LFG that should be destroyed. In this situation:

$F_{CH4,BL,y}=F_{CH4,BL,R,y}$

There is no requirement specifying the amount or percentage of LFG that should be destroyed, and no requirement on the installation of a system to capture and flare the LFG. As a result, it is assumed a typical destruction rate of 20 per cent by Option 2 (ii) to calculate the required LFG destruction.

$F_{CH4,BL,R,y}=\rho_{reg,y}\times F_{CH4,PJ,capt,y}$

Where:

$F_{CH4,BL,R,y}$	=	*Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (tCH_4/yr)*
$\rho_{reg,y}$	=	*Fraction of LFG that is required to be flared due to a requirement in year y. A default value of 20% may be used.*
$F_{CH4,PJ,capt,y}$	=	*Amount of methane in the LFG which is captured in the project activity in year y (tCH_4/yr)*

<u>Baseline emissions are determined according to the following equation for electricity part:</u>

$BE_{y,2}=EG_{PJ,y}\times EF_{grid,y}$

$BE_{y,2}$	=	*Baseline emissions associated with electricity generation in year y (tCO_2)*

$EG_{PJ,y}$	=	*Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)*
$EF_{Grid,y}$	=	*Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)*

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$EG_{PJ,y}=EG_{PJ,facility,y}$

Where:

$EG_{PJ,facility,y}$	=	*Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)*

Determination of the emission factor for electricity generation ($EF_{Grid,y}$)

The baseline scenario of the project is that the LFG from Lichuan landfill site was released directly into atmosphere and equivalent electricity generation by the project was supplied by CCPG, Scenario A of the "Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" applies. $EF_{Grid,y}$ shall therefore be determined in accordance with Option A1 of the tool, i.e. the applied emission factor shall be the combined margin emission factor of the CCPG, calculated in accordance with the "Tool to calculate the emission factor of an electricity system" ($EF_{Grid,y} = EF_{grid,CM,y}$).

The grid emission factor is calculated as the weighted average of the operating margin (0.5) & build margin (0.5) values. The value of combined margin is sourced from 2023 Baseline Emission Factors for Regional Power Grids in China dated 08/07/2024 published by China DNA. China DNA calculates the data based on Tool to Calculate the Emission Factor for an Electricity System", version 07.0 which is the latest version available during the validation process. No further assessment is required for grid emission calculation as the ex-ante value is sourced directly from the Chinese DNA.

Emission factor ($EF_{Grid,y}$): $EF_{Grid,y} = EF_{grid,CM,y} = 0.8771 \times 0.5 + 0.2696 \times 0.5 = 0.57335$ tCO_2/MWh.

This value is fixed ex-ante for the crediting period.

ID number	$F_{CH4,PJ,y}$[1]	$F_{CH4,BL,y}$[2]	$BE_{y,1}$

[1] This parameter is determined based on $BE_{CH4,SWDS,y}$, GWP_{CH4} and η_{PJ}, also considering the operation days of each year. Please refer to ER sheet for details.

[2] This parameter is determined based on $F_{CH4,PJ,y}$ and $\rho_{reg,y}$, please refer to ER sheet for details.

Unit	tCH_4	tCH_4	tCO_2e
05/09/2024 – 31/12/2024	703	141	16,152
2025	2,338	468	53,688
2026	2,500	500	57,405
2027	2,229	446	51,172
2028	1,996	399	45,830
2029	1,796	359	41,238
2030	1,624	325	37,276
2031	1,474	295	33,846
2032	1,344	269	30,865
2033	1,231	246	28,263
01/01/2034 – 04/09/2034	766	153	17,583

ID number	$EG_{PJ,y}$	$EF_{grid,CM,y}$	$BE_{y,2}$
Unit	MWh	tCO_2e/MWh	tCO_2e
05/09/2024 – 31/12/2024	2,957	0.57335	1,695
2025	9,829	0.57335	5,635
2026	10,510	0.57335	6,025
2027	9,368	0.57335	5,371
2028	8,391	0.57335	4,810
2029	7,550	0.57335	4,328
2030	6,825	0.57335	3,912
2031	6,197	0.57335	3,552
2032	5,651	0.57335	3,239
2033	5,174	0.57335	2,966

01/01/2034 – 04/09/2034	*3,219*	*0.57335*	*1,845*

Year	*Emission Reduction by Methane recovery from the landfill in the absence of the project activity during the crediting period (tCO_2e) ($BE_{y,1}$)*	*Emission Reduction from Electricity Generation (tCO_2e) ($BE_{y,2}$)*	*Baseline Emissions (tCO_2e)*
05/09/2024 – 31/12/2024	*16,152*	*1,695*	*17,847*
2025	*53,688*	*5,635*	*59,323*
2026	*57,405*	*6,025*	*63,430*
2027	*51,172*	*5,371*	*56,543*
2028	*45,830*	*4,810*	*50,640*
2029	*41,238*	*4,328*	*45,566*
2030	*37,276*	*3,912*	*41,188*
2031	*33,846*	*3,552*	*37,398*
2032	*30,865*	*3,239*	*34,104*
2033	*28,263*	*2,966*	*31,229*
01/01/2034 – 04/09/2034	*17,583*	*1,845*	*19,428*
Total (tones of CO_2e)	***413,318***	***43,378***	***456,696***

Project emissions

As per AMS-III.G, Project emissions are calculated as follows:

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y}$$

Where:

PE_y	=	*Project emissions in year y (tCO_2/yr)*
$PE_{power,y}$	=	*Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (tCO_2e)*
$PE_{flare,y}$	=	*Emissions from flaring or combustion of the landfill gas stream in the year y (tCO_2e)*
$PE_{process,y}$	=	*Emissions from the landfill gas upgrading process in the year y (tCO_2e), determined by following the relevant procedures described in annex 1 of AMS-III.H.*

According to AMS-III.G, project emissions from electricity consumption are determined as per the procedures described in AMS-I.D “Grid connected renewable electricity generation”. According to AMS-I.D, the project is neither involved geothermal power plants nor hydro power plants, project emissions from electricity consumption are identified as 0. The project does not involve the emissions from the consumption of fossil fuels, only emissions from the use of electricity for the operation of the installed facilities are involved. The project will consume electricity delivered from the grid when the project is not in operation. And this part of electricity has already been deducted from the quantity of electricity supplied by the project plant/unit to the grid when calculating the Quantity of net electricity generation supplied by the project plant/unit to the grid, which is then used to calculate the Baseline emissions associated with electricity generation ($BE_{EC,y}$) hence emissions from the use of electricity for the operation of the installed facilities are excluded, which means Emissions from the use of fossil fuel or electricity for the operation of the installed facilities equal to 0. Therefore, $PE_{power,y}$ is equal to 0 for ex-ante estimation.

There is no flare used to destroy the LFG, therefore, $PE_{flare,y}$ Is equal to 0.

The project captures LFG for power generation and not involved upgrading process, therefore, $PE_{process,y}$ is equal to 0.

Therefore, PE_y = 0 tCO_2e

Leakage

As the methane recovery technology is not equipment transferred from another activity, leakage effects are to be considered. The validation team deems this consideration is correct and in line with methodology AMS-III.G, version 10.0 and AMS-I.D, version 18.0.

Therefore, LE_y = 0 tCO_2e

Emission Reductions

According to AMS-III.G, version 10.0 and AMS-I.D, version 18.0, emission reductions are calculated as follows:

$ER_y = BE_y - PE_y - LE_y$

Where:

ER_y	=	*Emission reductions in year y (tCO_2e/yr)*
BE_y	=	*Baseline emissions in year y (tCO_2e/yr)*
PE_y	=	*Project emissions in year y (tCO_2e/yr)*
LE_y	=	*Leakage in year y (tCO_2e/yr)*

Hence for the project activity, the estimated amount of GHG emission reductions (ER_y) is 456,696 tCO_2e during the crediting period from 05/09/2024 to 04/09/2034, resulting in estimated average annual emission reductions of 45,669 tCO_2e.

Year	***Estimated baseline emissions or removals (tCO_2e)***	***Estimated project emissions or removals (tCO_2e)***	***Estimated leakage emissions (tCO_2e)***	***Estimated net GHG emission reductions or removals (tCO_2e)***
05/09/2024 – 31/12/2024	*17,847*	*0*	*0*	*17,847*
2025	*59,323*	*0*	*0*	*59,323*
2026	*63,430*	*0*	*0*	*63,430*
2027	*56,543*	*0*	*0*	*56,543*
2028	*50,640*	*0*	*0*	*50,640*
2029	*45,566*	*0*	*0*	*45,566*
2030	*41,188*	*0*	*0*	*41,188*
2031	*37,398*	*0*	*0*	*37,398*
2032	*34,104*	*0*	*0*	*34,104*
2033	*31,229*	*0*	*0*	*31,229*
01/01/2034 – 04/09/2034	*19,428*	*0*	*0*	*19,428*
Total	***456,696***	***0***	***0***	***456,696***

3.3.7 Methodology Deviations

The equation (6) in the applied methodology AMS-III.G is wrong as additional GWP_{CH4} is multiplied. Therefore, during the validation process, a correction was made in the PD for ER calculation without considering GWP_{CH4} as equation (22).

The assessment team consider the deviation for methodology is correct applied due the error in the methodology. This deviation would not negatively impact the conservativeness of the quantification of GHG emission reductions or carbon dioxide removals, and do not relate to any other part of the methodology.

3.3.8 Monitoring Plan

Through the site visit and checking related documents, the assessment team confirms that ex-ante parameter has been correctly presented in the MR and in line with the registered PD and requirement of methodology and tools.

Parameters determined ex-ante

Data / Parameter:	OX
Data unit:	/
Description:	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data used:	Consistent with an extensive review of published literature on this subject, including the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.1

Data / Parameter:	GWP_{CH4}
Data unit:	$tCO_2e/t\ CH_4$
Description:	Global warming potential of CH_4
Source of data used:	IPCC Fifth Assessment Report (AR5)
Value applied:	28

Data / Parameter:	ρ_{CH4}
Data unit:	t/m^3

Description:	*Density of methane gas at Normal Conditions*
Source of data used:	*Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0*
Value applied:	*0.0007156*

Data / Parameter:	η_{PJ}
Data unit:	*Dimensionless*
Description:	*Efficiency of the LFG capture system that will be installed in the project activity*
Source of data used:	*FSR of the project (based on technical specifications of the LFG capture system)*
Value applied:	*85%*

Data / Parameter:	φ_y
Data unit:	*Dimensionless*
Description:	*The model correction factor to account for model uncertainties*
Source of data used:	*Default value of the tool "Emissions from solid waste disposal sites" (version 08.1)*
Value applied:	*0.75*

Data / Parameter:	f_y
Data unit:	*Dimensionless*
Description:	*Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y*
Source of data used:	*As in the baseline scenario, there is no LFG capturing and usage, zero is considered and in line with the requirement of tool "Emissions from solid waste disposal sites" (version 08.1)*
Value applied:	*0*

Data / Parameter:	F
Data unit:	/
Description:	Fraction of methane in the SWDS gas (volume fraction)
Source of data used:	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5

Data / Parameter:	$DOC_{f,y}$
Data unit:	Weight fraction
Description:	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data used:	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5

Data / Parameter:	MCF_y
Data unit:	/
Description:	Methane correction factor
Source of data used:	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	1.0 Lichuan landfill site is an anaerobic managed solid waste disposal site. As confirmed through site visit and checking FSR of the project, Lichuan landfill site has controlled placement of waste by mechanical compacting and levelling of the waste, then value 1.0 has been applied

Data / Parameter:	DOC_j
Data unit:	/
Description:	Fraction of degradable organic carbon in the waste type j (weight fraction)

Source of data used:	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	For MSW, the following values for the different waste types j should be applied: <table><tr><td>Waste type j</td><td>DOC_j (% wet waste)</td></tr><tr><td>Wood and wood products</td><td>43</td></tr><tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr><tr><td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr><tr><td>Textiles</td><td>24</td></tr><tr><td>Garden, yard and park waste</td><td>20</td></tr><tr><td>Glass, plastic, metal, other inert waste</td><td>0</td></tr></table>

Data / Parameter:	k_j
Data unit:	1/yr
Description:	Decay rate for the waste type j
Source of data used:	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	Apply the following default values for the different waste types j: <table><tr><td colspan="2">Waste type j</td><td>k_j (MAT≤20℃, MAP/PET>1)</td></tr><tr><td rowspan="2">Slowly degrading</td><td>Pulp, paper, cardboard (other than sludge)</td><td>0.06</td></tr><tr><td>Wood, wood products and straw</td><td>0.03</td></tr><tr><td>Moderately degrading</td><td>Other (non-food) organic putrescible garden and park waste</td><td>0.10</td></tr></table>

	<table><tr><td>Rapidly degrading</td><td>Food, food waste, sewage sludge, beverages and tabacco</td><td>0.185</td></tr></table> *As per the FSR of the project activity and online information, the climate data for the location of Lichuan landfill site is:* *Temperature: 13.2 ℃* *MAP – Mean annual precipitation: 1,400 mm* *PET – Potential evapotranspiration: 1,079.4 mm*

Data / Parameter:	$W_{j,x}$
Data unit:	*t*
Description:	*Amount of organic waste type j disposed/prevented from disposal in the SWDS in the year y*
Source of data used:	*Estimated once in the FSR* *The total amount of waste is sourced from FSR based on historic data and forecast for the future situation. Considering FSR was compiled by third-party with accreditation and approved by government as project approval. FSR is considered as a reliable source for data.*
Value applied:	*Apply the following default values for the different waste types j:*

Year	***W_j (waste amount) (t/x)***
2007	*5,873*
2008	*72,916*
2009	*74,404*
2010	*75,923*
2011	*76,690*
2012	*77,464*
2013	*78,247*
2014	*79,037*
2015	*79,836*
2016	*80,642*
2017	*81,457*
2018	*82,279*
2019	*83,111*
2020	*83,950*
2021	*125,208*

	2022	126,473
	2023	127,750
	2024	134,776
	2025	142,189
	2026	150,009
	Waste type of Landfill site	**Weight content （%）**
	W_1-Wood and wood products	12.95
	W_2-Pulp, paper and cardboard	12.90
	W_3-Food, food waste, beverages and tobacco	37.36
	W_4-Textiles	1.57
	W_5-Garden, yard and park waste	1.00
	W_6-Glass, plastic, metal other inert	34.22

Data / Parameter:	$EF_{grid,OM,y}$
Data unit:	tCO_2/MWh
Description:	Operating margin CO_2 emission factor in year y
Source of data used:	2023 Baseline Emission Factors for Regional Power Grids in China dated 08/07/2024 published by China DNA
Value applied:	0.8771

Data / Parameter:	$EF_{grid,BM,y}$
Data unit:	tCO_2/MWh
Description:	Build margin CO_2 emission factor in year y
Source of data used:	2023 Baseline Emission Factors for Regional Power Grids in China dated 08/07/2024 published by China DNA
Value applied:	0.2696

Data / Parameter:	NCV_{CH4}

Data unit:	*MJ/Nm³*
Description:	*Net calorific value of methane at reference conditions*
Source of data used:	*AMS-III.G*
Value applied:	*35.9*

Parameters determined ex-post

- *Policy requirement: $F_{CH4,BL,y}$, $\rho_{reg,y}$*

Parameters title	**Descriptions**
$F_{CH4,BL,y}$	*Amount of methane in the LFG which is flared due to a requirement in year y*
$\rho_{reg,y}$	*Fraction of LFG that is required to be flared due to a requirement in year y*

$F_{CH4,BL,y}$ and $\rho_{reg,y}$ will be checked annually through information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odour concerns. The process, measurement methods, procedures and calculation of $F_{CH4,BL,y}$ has been presented in the 3.3.6 of the report and the applied value is determined as 20% as case 2 was selected.

- *Electricity Generation: $EG_{PJ,facility,y}$, EG_y*

Parameters title	**Descriptions**
$EG_{PJ,facility,y}$	*Quantity of net electricity generated supplied by the project plant/unit in year y*
EG_y	*Electricity generation in year y*

$EG_{PJ,facility,y}$ and EG_y will be continuous measured by electricity meters with accuracy of 0.5s and monthly recorded. The electricity meters will be calibrated in accordance with national standards or other relevant requirements. The accuracy of electricity meters will be satisfied with relevant standards or requirements. The type, serial number, location, function and accuracy of the electricity meters have been confirmed through site visit.

- *LFG utilization: $D_{CH4,y}$, P, T and EE_y*

Parameters title	**Descriptions**

$D_{CH4,y}$	*Density of methane at the temperature and pressure of the landfill gas in year y*
P	*Pressure of the landfill gas*
T	*Temperature of the landfill gas*
EE_y	*Energy Conversion Efficiency of the project equipment*

$D_{CH4,y}$ was supposed to be calculated by based on T and P. However, according to the project owner and site visit, the project will employ landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas will be displayed.

P was supposed to be monitored by manometer. However, according to the project owner and site visit, the project will employ landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas will be displayed.

T was supposed to be monitored by thermometer. However, according to the project owner and site visit, the project will employ landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas will be displayed.

EE_y will be determined according to paragraph 21 of AMS-III.G. (version 10.0), specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for biogas. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation.

Through the site visit, the validation team confirm the correctness of following information:

a) Methods for measuring, recording, storing, aggregating, collating and reporting data and parameters;

b) Policies for oversight and accountability;

c) Procedures for internal auditing and QA/QC;

d) Procedures for handling non-conformances with validated monitoring plan.

By site visit, it is confirmed that manufacturer details of the monitoring equipment presented in the PD is accurate and in line with the actual situation.

By site visit and checking related regulations, it is confirmed that the calibration period of the monitoring equipment presented in the PD is accurate and in line with the requirements in related regulations.

By site visit, it is confirmed that the training and maintenance procedures and emergency procedures for monitoring system is in place and in line with the description in the PD.

Based on the on-site visit and interviewed with the Manager of Lichuan City BCCY New Energy Co., Ltd., the validation team confirmed that monitoring parameter (including ex-ante parameters and ex-post parameters) has been correctly descried in the PD and in compliance with the methodology AMS-III.G Version 10.0, AMS-I.D Version 18.0 and the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and VCS via VCS Standard version 4.7. Also, the monitoring equipment and location has been confirmed through the site visit and are in line with the requirements of methodology.

In conclusion, the assessment team confirmed that all parameters those fixed at validation and those that need to be monitored in line with applied methodology are included in the monitoring plan and in line with methodology.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION OPINION

4.1 Validation Summary

Applus+ Certification has been engaged by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project (VCS Ref. No. 5313).

The management of the project proponent/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project's Monitoring Plan and the approved methodology AMS-III.G, version 10.0 and AMS-I.D, version 18.0.

4.2 Validation Conclusion

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

The project's description compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide version 4.4 and VCS standard version 4.7.

The project's baseline and additionality and monitoring plan are assessed against AMS-III.G, version 10.0 and AMS-I.D, version 18.0 for small scale project.

A risk based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Centre S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

The data and information supporting the GHG statement assertion were hypothetical, projected and/or historical in nature. The reasonableness of assumptions, limitations, and methods that support a claim about the outcome of future activities, explaining that actual results may vary since the estimates are based on assumptions that are subject to change. The project is likely to achieve the estimated GHG emission reductions and carbon dioxide removals described below.

Crediting Period: From 05/09/2024 to 04/09/2034

Validated estimated GHG emission reductions and carbon dioxide removals for the project

crediting period:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
05/09/2024 – 31/12/2024	17,847	0	0	17,847	0	17,847
2025	59,323	0	0	59,323	0	59,323
2026	63,430	0	0	63,430	0	63,430
2027	56,543	0	0	56,543	0	56,543
2028	50,640	0	0	50,640	0	50,640
2029	45,566	0	0	45,566	0	45,566
2030	41,188	0	0	41,188	0	41,188
2031	37,398	0	0	37,398	0	37,398
2032	34,104	0	0	34,104	0	34,104
2033	31,229	0	0	31,229	0	31,229
01/01/2034–04/09/2034	19,428	0	0	19,428	0	19,428
Total	456,696	0	0	456,696	0	456,696

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification	Assessment method and conclusion
Not applicable	*Not applicable*	*Not applicable*	*Not applicable*

APPENDIX II: REFERENCE LIST

1. *PD version 01 dated 21/10/2024, version 04 dated 24/11/2025*
2. *ER calculation spreadsheet for ex-ante estimation*
3. *Board resolution dated 20/03/2024*
4. *Approved methodology AMS-III.G: Landfill methane recovery, version 10.0*

   Approved methodology AMS-I.D: Grid connected renewable electricity generation, version 18.0
5. *CDM Validation and Verification Standard for project activities, version 03.0*
6. *CDM Project Standard for project activities, version 03.0*
7. *Tool to calculate the emission factor for an electricity system, version 07.0*
8. *Emissions from solid waste disposal sites, version 08.1*
9. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*
10. *Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0*
11. *Positive lists of technologies, version 04.0*
12. *VCS standard version 4.7, dated 16/04/2024*
13. *VCS Program Guide version 4.4, dated 17/01/2023*
14. *Business License of Lichuan City BCCY New Energy Co., Ltd.*
15. *Feasibility Study Report dated 01/2024*
16. *Project Approval dated 27/02/2024*
17. *Environmental Impact Assessment Report dated 04/2024*
18. *Environmental Impact Assessment (EIA) Approval dated 18/04/2024*
19. *Nameplate of the equipment*
20. *Construction Order dated 20/04/2024*
21. *Stakeholder Meeting Minutes dated 19/04/2024*
22. *Local stakeholder consultation questionnaire*
23. *Stakeholder Meeting Notice dated 18/03/2024*
24. *Grievance Expression Book located in the project site*
25. *Operation log*
26. *Cooperation agreement between Lichuan City Urban Management Law Enforcement Bureau and Henan BCCY Environmental Energy Co., Ltd. dated 26/12/2023*

27. *2023 Baseline Emission Factors for Regional Power Grids in China dated 08/07/2024*

28. *Technical specification of equipment*

29. *Declaration for no double counting issued by project proponent*

30. *"Notice on Key Work Related to the Management of Greenhouse Gas Emission Reports for Enterprises in 2022" issued by Ministry of Ecology and Environment of P. R. China*

31. *REC Mechanism database of China: https://www.greenenergy.org.cn/*

32. *I-REC website: https://www.irecstandard.org*

33. *Regulation and laws:*

"Women's Rights Protection Law"

http://www.scio.gov.cn/ztk/xwfb/46/11/Document/976078/976078_3.htm

"Minor Protection Law"

https://www.gov.cn/xinwen/2020-10/18/content_5552113.htm

"Labor Law"

https://www.gov.cn/banshi/2005-05/25/content_905.htm

"Criminal Law"

http://gongbao.court.gov.cn/details/f8e30d0689b23f57bfc782d21035c3.html

34. *HR Records and Staff Roaster*

35. *VCS monitoring manual*

36. *Training Records*

37. *"Standard for Pollution Control on the Landfill Site of Municipal Solid Waste" (GB 16889-2008)*

38. *"Technical Code for Municipal Solid Waste Sanitary Landfill" (GB 50869-2013)*

39. *Land certificate*

40. *2021 Urban Construction Statistical Yearbook*

41. *2021 Urban and Rural Construction Statistical Yearbook*

42. *Report on Methane Emission Control and Resource Utilization*

43. *Venue lease contract between Lichuan City Environmental Sanitation Service Center and Lichuan City BCCY New Energy Co., Ltd. dated 18/01/2024*

APPENDIX III: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

CL ID	01	Section no.	3.3.2	**Date:** 01/01/2025
Description of CL				
Please clarify why the project activity is applicable for small-scale methodology considering the estimated emission reductions for the project in 2026 is over 60,000 tCO_2e.				
Project proponent response				**Date:** 04/01/2025
According to paragraph 4 of AMS-III.G (version 10.0), emission reductions are required less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity. In this project, the emission reduction from methane destroy is 57,405 tCO_2 in 2026, which is less than 60 kt CO_2. Therefore, small scale meth is applicable to the project.				
Documentation provided by project proponent				
Updated PD				
VVB assessment				**Date:** 05/01/2025
By checking updated PD, it is confirmed that the emission reduction from methane destroy is 57,405 tCO_2 in 2026, which is less than 60 kt CO_2, therefore, small-scale methodology is still applicable. Therefore, CL01 was closed.				